BDA400 | DEVELOPMENT PHASE

Technical Analysis Using R

Assignment 5

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Course	BDA400 - Data Science Tools and Techniques
Project	Base-R technical indicators, filters, and trading rule
Repository	github.com/lubnaafandi89-coder/TechnicalAnalysis

Project purpose

This project implements the required technical-analysis indicators in separate R scripts using core R calculations. The included runner applies them to NVDA, AMZN, and TSLA input data and writes analysis and signal CSV files.

Visual key: blue headings and red section accents identify this report.

Run the project in RStudio

Folder layout

Keep **Assignment2** and **Assignment5** together at the repository root. Set the working directory to Assignment5 before running the scripts.

Step	Action
1	Run source("00_prepare_inputs.R"). It uses quantmod for data import only and saves daily OHLCV CSVs for NVDA, AMZN, and TSLA in ../Assignment2/outputs/.
2	Run source("99_validate_functions.R") to check the base-R indicator functions.
3	Run source("00_run_assignment5.R") to calculate indicators and export results.
4	Review the Console summary and CSV files in Assignment5/outputs/.

Files created

Per-symbol technical analysis: ***_technical_analysis.csv**; filtered observations: ***_filtered_signals.csv**; combined latest-value summary: **assignment5_summary.csv**. Results are written to Assignment5/outputs/. No indicator package is used; quantmod is only for importing OHLCV data.

Implementation overview

Each indicator is in its own file under functions/. Function names follow the assignment templates.

Script	Implementation summary
01_sma.R	Simple moving average from complete windows.
02_ema.R	Recursive exponential average using the specified smoothing factor.
03_macd.R	Short EMA minus long EMA, signal EMA, and histogram.
04_stdev.R	Standard deviation using population variance (divide by n).
05_linreg.R	Linear regression coefficients and projected values.
06_rsi.R	Relative Strength Index with Wilder-style average gain and loss.
07_stoch_rsi.R	Rolling RSI normalization followed by smoothed %K and %D.
08_crossover.R	Detects upward crossings between aligned series.
09_crossunder.R	Detects downward crossings between aligned series.
10_filters_and_trading_rules.R	Example recent-date/close-above-SMA filter and long/flat SMA20-SMA50 rule.

Alignment notes

The runner pads early SMA rows with NA to preserve date alignment. The example filter keeps the latest 90 calendar days where Close is above SMA20. The position rule enters Long at an upward SMA20/SMA50 cross and returns to Flat at a downward cross.

Submission checklist

Upload the Assignment5 folder to the public GitHub repository. Submit the explanatory PDF to the LMS using the instructor-requested filename. Keep the PDF and code together in the repository for review.